HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Supported technical activities for clinical and commercial manufacturing operations, contributing to process monitoring and trend analysis for **pharmaceutical products**.
- Assisted with **investigations, deviations, CAPAs, and change controls** by performing scientific and technical assessments within a regulated **GMP environment**.
- Contributed to process optimisation initiatives aimed at improving **quality, compliance, and efficiency** by analysing manufacturing and process performance data.
- Collaborated with cross-functional teams including Manufacturing, Quality, and Supply Chain to resolve manufacturing and **technical issues**.
- Supported the preparation and review of technical documentation such as protocols, reports, and **process descriptions**.
- Assisted in equipment qualification and the implementation of new technologies and manufacturing capabilities, focusing on **process scale-up activities**.
- Containerized and deployed Python-based data processing applications to **AWS Lambda**, enabling event-driven execution for **data analysis** tasks.

SKILLS

Pharmaceutical Science – Process Engineering, Technology Transfer, Process Validation, Continuous Improvement, GMP Environment

Data Analysis Tools – Power BI, Tableau, R, Python (pandas, NumPy, scikit-learn), SQL

Software Development – Java, Spring Boot, Python, REST APIs, Docker, Git, Jenkins

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Database Technologies – PostgreSQL, MySQL, Oracle

Enterprise Systems – ERP Integration, Change Management, Splunk, Dynatrace

PROJECTS

Process Improvement for Manufacturing Data Analysis

- Developed Python scripts to automate the extraction and analysis of manufacturing data from diverse sources, identifying key performance indicators for process optimisation.
- Built interactive **Power BI dashboards** to visualise process trends, equipment performance, and quality metrics, enabling data-informed decision-making for operations teams.
- Applied statistical methods to analyse process variation, contributing to a **15% reduction** in batch reprocessing incidents over a 6-month period.

Automated Production Monitoring and Alerting System

- Designed and implemented a Python-based system to monitor production line status and critical process parameters in near real-time.
- Integrated with **Splunk** for log aggregation and developed custom alerts for deviations, reducing manual oversight by **~40%** for the operations team.
- Developed REST APIs using **Flask** to expose monitoring data for integration with enterprise reporting tools, supporting **SLA management** for key manufacturing workflows.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)